

Securitize BUIDL-I Deploy Diff Audit Summary

Prepared by Cyfrin

Version 1.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on BUIDL-I Deploy Diff. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number of findings by severity. It closes with Cyfrin's attestation of the review's scope.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 1.0 dated September 9, 2026, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

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5 Audit Overview

Over the course of 2 days, the Cyfrin team conducted an audit on the BUIDL-I Deploy Diff code provided by Securitize.

Summary

Project Name	BUIDL-I Deploy Diff
Repository	dstoken
Commit	d78d34a3ec41...
Audit Timeline	Sep 7th - Sep 8th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	–	–
High Risk	0	–	–
Medium Risk	0	–	–
Low Risk	0	–	–
Informational	0	–	–
Gas Optimizations	0	–	–
Total	0	–	–

The report records no findings at any severity, so no statuses are shown.

6 Audit Scope

The review covered the range `262370ae..d78d34a3` of the securitize-io/dstoken repository against the live BUIDL-I deployment on Ethereum mainnet (token proxy `0x6a9da2d710bb9b700acde7cb81f10f1ff8c89041`):

- Audited baseline: 262370ae848bde6bf0b5d91e6ff90ef94768709d
- Review target: d78d34a3ec41ab9e139670aeeb7763133215b34c, named in the request as the head of the `master` branch

The audit scope was limited to reviewing the Ethereum mainnet on-chain state of the BUIDL-I smart contracts to determine:

- which commit(s) the deployed BUIDL-I smart contracts match
- which, if any, of the following commits have a security impact on the deployed BUIDL-I smart contracts:

Commit	Date	Summary
`e07f193f`	2024-08-07	bulk operator
`ec720473`	2024-08-07	swap
`27a64559`	2024-08-08	swap test

Commit	Date	Summary
705b59b0	2024-08-08	swap test migrated completely
61022c93	2024-08-08	npmrc and console.log cleanup
ff3c53bf	2024-08-08	changeset / publish commands
efffddc5	2024-08-08	pipeline update
8e8eb4c7	2024-08-09	package.json version 3.1.1
580ab1c1	2024-08-13	added gap pattern
03d673ca	2024-08-14	merge (PR #164)
d6424fff	2024-10-15	update transaction relayer
1ef40806	2024-10-17	add getComplianceTransferableToken in compliance interface
440eb4f9	2024-10-17	bump version 3.1.2
d8356d27	2024-10-17	merge, complete compliance interface (PR #166)
63c800a3	2024-10-18	preserve interface in CS for backward compatibility
d7ba5f6a	2024-10-23	gitignore zksync artifacts
366f8515	2024-10-23	merge (PR #167)
fac63e9a	2024-10-31	SPDX GPL-3.0 to Apache-2.0, add disclaimer
851a4f37	2024-11-01	adhoc deploy libraries task
9bb8ad15	2024-11-01	fix loggers
5ead44c6	2024-11-03	merge (PR #169)
d78d34a3	2024-11-03	merge dev (PR #170), the review target

The source review covered the components changed in that range:

- `BulkOperator` and `IBulkOperator`
- `TransactionRelayer`
- `IDSComplianceService`, `ComplianceServiceRegulated`, inherited `ComplianceServiceWhitelisted` code and `ComplianceServiceLibrary`
- deployment configuration: the `BulkOperator` deployment task, service registration, role setup, the library deployment task and service constants

Where a changed path called into `DSToken`, `TrustService`, `RegistryService`, `TokenIssuer`, `WalletRegistrar`, `WalletManager`, `InvestorLockManager`, `ComplianceConfigurationService`, `OmnibusTBEController`, `TokenReallocator`, `IssuerMulticall`, `TokenLibrary` or shared authorization code, those were reviewed as dependencies.

This is a diff review of that range against the live BUIDL-I deployment: matching bytecode does not mean unchanged baseline code was re-audited. Later work on `master` is out of scope.

7 Deployment Verification

The review identified the BUIDL-I deployment on Ethereum mainnet and compared its deployed code with review target commit d78d34a3ec41ab9e139670aeeb7763133215b34c of securitize-io/dstoken.

7.1 Deployment identification

Cyfrin scanned Ethereum mainnet from 1 October 2024 through 7 September 2026 for `DSService-Set(uint256,address)` events with no address filter: 858 logs from 225 emitters. Of the token candidates checked through `name()` and `symbol()`, only one returned BUIDL-I:

```
name:   BlackRock USD Institutional Digital Liquidity Fund - I Class
symbol: BUIDL-I
proxy:  0x6a9da2d710bb9b700acde7cb81f10f1ff8c89041
```

The token's `DSServiceSet` history and current `getDSService(id)` values show 12 non-zero services. The scope is the token, those 12 services and IssuerMulticall: 14 proxies, their 14 implementations and 2 linked libraries, 30 deployed objects in total. Library addresses are from `settings.libraries` in the verified Etherscan compiler inputs; local link references and the deployed consuming bytecode match those addresses.

7.2 Covered proxies, implementations and libraries

```
DSToken
  proxy:          0x6a9da2d710bb9b700acde7cb81f10f1ff8c89041
  implementation: 0x9e2693f54831f6f52b0bb952c2935d26919a3626
TrustService (service ID 1)
  proxy:          0x712244c054244432d0f9ea16a56121598eba7a2c
  implementation: 0x54687c2bd1bcb739bc2e9f45ae8d9ab27b9a5623
RegistryService (service ID 4)
  proxy:          0xf8e91fa34311876302d36d14b4f246044fd4332a
  implementation: 0x9aa37ef5b1e75e4c6f7e87f79ac2a4d06e23748a
ComplianceServiceRegulated (service ID 8)
  proxy:          0xc7ca0ce7626940838a87656656c6af92b95abd00
  implementation: 0xb2241faa9e359d131f73537999d6579b95c04c34
WalletManager (service ID 32)
  proxy:          0x1df0616fd703643411d62c34dfbfa3a261df7dab
  implementation: 0x487bfff0719a3a66a699b3388c103d2d4e4af748
InvestorLockManager (service ID 64)
  proxy:          0x32708a900a740846445fd49c0c00df15701720a2
  implementation: 0xea9b5ad94fea52e092f518cd09bff35d75c9edcc
ComplianceConfigurationService (service ID 256)
  proxy:          0x41a5f31c45fa12913e125914bd79ef49befa1a1f
  implementation: 0xbe729b88ba6f29fa06a76b8d7c87e11730bf8bf8
TokenIssuer (service ID 512)
  proxy:          0xd8305ccd41fc67bd7fc0d6653a89452f4c1f7265
  implementation: 0x12af61ccbceffad492d749f97d22e2af9d7b1ee1
WalletRegistrar (service ID 1024)
  proxy:          0x51d016b8cc89f56fd5a7970892586fb512e202a0
  implementation: 0x26c1b0c7cb630e1b69cc9796021958a911cef470
OmnibusTBEController (service ID 2048)
  proxy:          0xddf7b95d79872766b4c0f32284fac86155c81b3a
  implementation: 0x8f3c283825cf9882e800a8465fd006d884d89bf3
TransactionRelayer (service ID 4096)
  proxy:          0xfd45dd1f78f9501b4e37b7a02e24ec29da64917f
  implementation: 0xa9dd4d527bb3974188a01522f888a2e38265bcc9
TokenReallocator (service ID 8192)
  proxy:          0x95e07b0682da8627decbb068cb57ae7ebe650e9e
  implementation: 0xa36c1c27df63198096366ce5f77811da0a641646
BulkOperator (service ID 8193)
  proxy:          0x619601b45bf5b2700ff10fa617bae7709ee63ff0
  implementation: 0xc71f2afd6239ee4d2275a5dd29b11426d0de3729
IssuerMulticall
  proxy:          0x34b3724b16db59fdd8b98cf3b3928100b010395b
  implementation: 0xa6e5f33b2372118ac70552c7212026d96af6f74b
TokenLibrary (linked into DSToken)
  library:        0x550bcacd69c722e19eee0599ef502c358ebc97e5
ComplianceServiceLibrary (linked into ComplianceServiceRegulated)
  library:        0x2a49ee732c76837b7e7e181e9c3c43946eed528c
```

Three contracts deployed after the review target (two DSTokenClassSwap proxies and SecuritizeBridge) have no source in the reviewed range and are outside this 30-object set.

7.3 Proxy history

Cyfrin queried `Upgraded(address)` for all 14 proxies from block zero through block 25,931,738. Each of the 14 proxies has exactly one `Upgraded` event, emitted in its creation transaction (14/14), and that event's implementation matches the current EIP-1967 implementation slot (14/14); no post-deployment upgrades were identified. The changes in the reviewed range that apply to BUIDL-I were therefore already present in the implementations installed when the proxies were created.

7.4 Build and bytecode comparison

Verified compiler inputs and build settings for all 30 addresses were taken from Etherscan. Source at `d78d34a3` was compiled with those settings and compared with each deployment transaction and with the live `eth_getCode`. Constructor and initializer payloads were decoded for all 14 proxies.

Compiler setting	Value
Solidity	0.8.20+commit.a1b79de6
Optimizer	Enabled, 200 runs
EVM version	paris
viaIR	false
OpenZeppelin contracts	5.0.2 from the target lockfile

Comparison	Result
Contract-deployment code, before constructor arguments and compiler metadata	30/30 match
Live contract code, excluding compiler metadata and expected deployment-specific address fields	30/30 match
Full proxy runtime bytecode	14/14 exact match
Implementation logic after filling the UUPS deployment address	14/14 match
Library logic after filling the library's deployed address	2/2 match
Proxy constructor and initializer decoding	14/14 successful
Unexplained contract-logic differences	0

The 14 implementations and 2 libraries differ in their IPFS metadata digests. Implementations also embed the UUPS implementation address, and each library embeds its own deployed address. These deployment-specific fields do not change contract logic. The comparison confirms the deployed runtime logic, but does not claim identical original-build metadata for those 16 objects.

All 14 proxies are `ERC1967Proxy(address _logic, bytes _data)` deployments. For each proxy, `_logic` is the implementation listed above and agrees with the creation-time `Upgraded` event and the current EIP-1967 implementation slot, and `_data` decodes against the initializer in the reviewed target ABI. All 14 creation receipts succeeded.

7.5 IssuerMulticall source qualification

The Etherscan-published source bundle for IssuerMulticall is not the same source snapshot as the review target: its 19 repository source files match commit `b5e71dcf0bcdd765dfb23b40df0dbf73807458e9` (11 June 2024) rather than `d78d34a3`. Both the published source and the target source reproduce the deployed IssuerMulticall runtime logic after accounting for the expected UUPS implementation-address fields and excluding Solidity metadata. Correspondence for IssuerMulticall therefore confirms runtime logic, not identical published source inputs or exact original-build provenance.

7.6 Result

Executable runtime correspondence was established for all 30 scoped objects against target commit `d78d34a3ec41ab9e139670aeeb7763133215b34c`. The 14 proxies are full byte-for-byte runtime matches. The 14 implementations and 2 libraries match in their executable regions after accounting for the documented deployment-specific addresses and excluding compiler metadata. No unexplained executable difference was identified. This conclusion is subject to the IssuerMulticall source qualification above and excludes the three contracts deployed after the review target.

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

DSToken is a rebasing ERC-20 security-token system that coordinates investor registration, transfer restrictions, compliance rules and issuance through a registry of supporting services. BUIDL-I is a DSToken deployment on Ethereum mainnet.

10 Attestation

Cyfrin attests that it performed a security review of the BUIDL-I Deploy Diff code provided by Securitize.

The repositories, commits, review period and findings are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-09-09-cyfrin-securitize-evm-buildi-deploy-diff-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=fdba05a6775c92d40890d0acc8f743dde62ed298f4d0a3b282389e677b15d88f
F=2026-09-09-cyfrin-securitize-evm-buildi-deploy-diff-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.